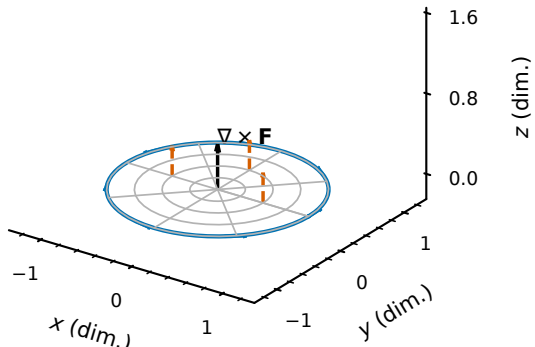
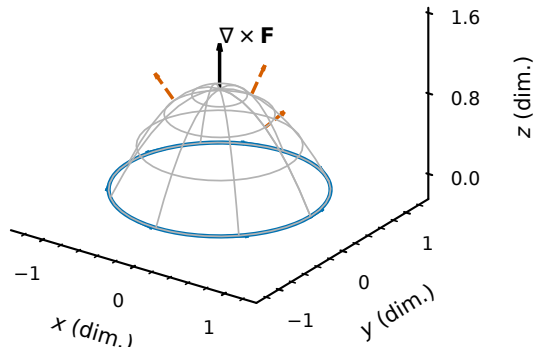
A



Disk: $\iint (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dS = 2\pi$

B



Cap $z = 1 - r^2$: same flux 2π